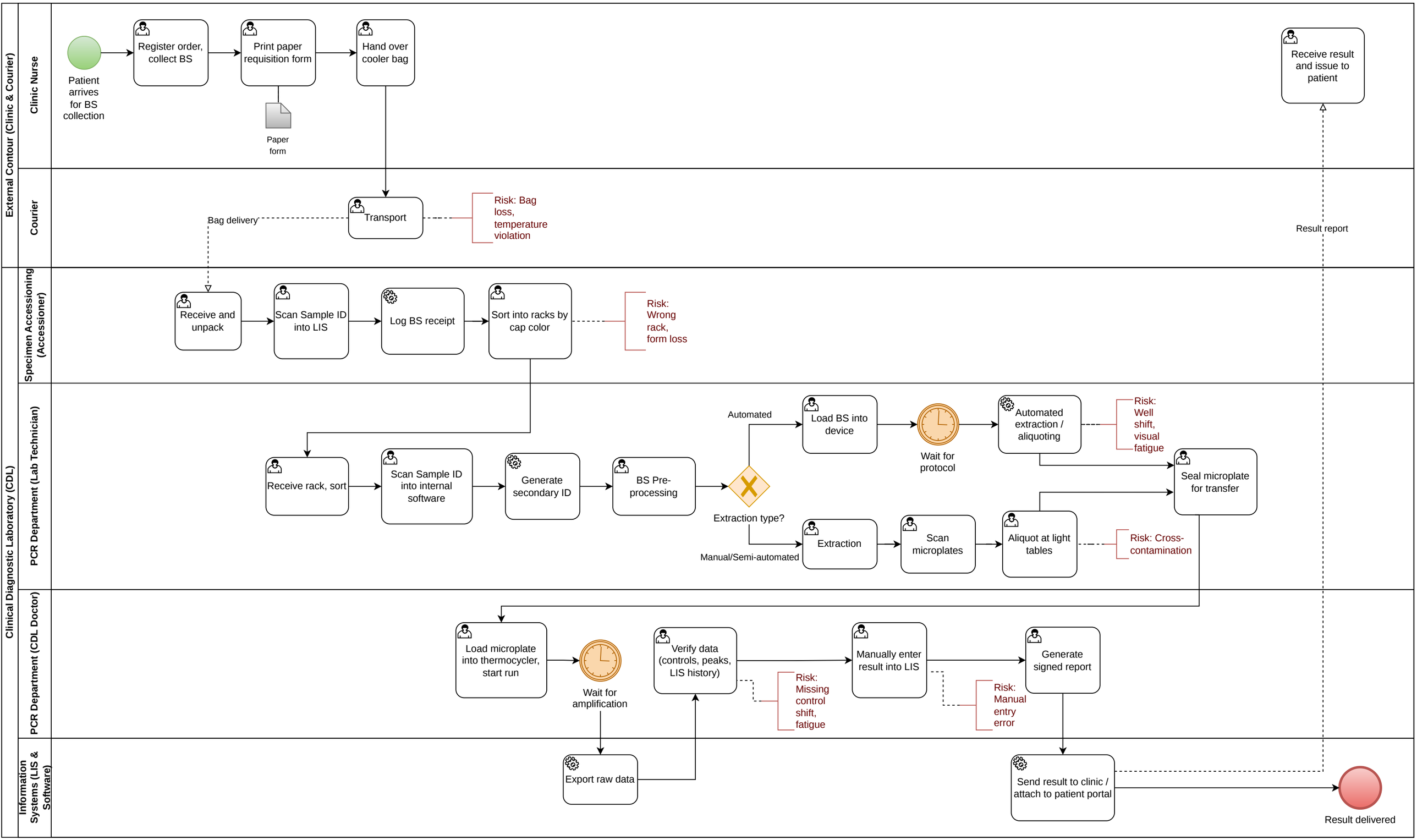
Business Process: PCR Testing — From Specimen Accessioning to Result Delivery

Notation: BPMN 2.0 • Participants: Clinic, Courier, CDL (Accessioning, Lab Tech, CDL Doctor), LIS • Version: 1.0 (Portfolio)



Legend (BPMN 2.0)

- Start Event
- End Event
- User Task
- Service Task
- Timer Intermediate Event
- Exclusive Gateway (XOR)
- Sequence Flow (solid line)
- Message Flow (between pools)
- Association (text link)
- Text Annotation (risks & bottlenecks)

Assumptions & Business Rules

- One Sample ID strictly corresponds to one patient
- BS is not transferred to amplification zone without a hermetically sealed microplate
- Manual result entry into LIS is allowed only after control verification by the PCR doctor.

Risk & Bottleneck Analysis

- **Transport:** Temperature violation, courier bag loss
- **Accessioning:** Paper requisition form loss
- **PCR Dept:** Cross-contamination with open wells, human factor (well shift, visual fatigue)
- **Validation:** Manual data entry error from thermocycler UI to LIS.